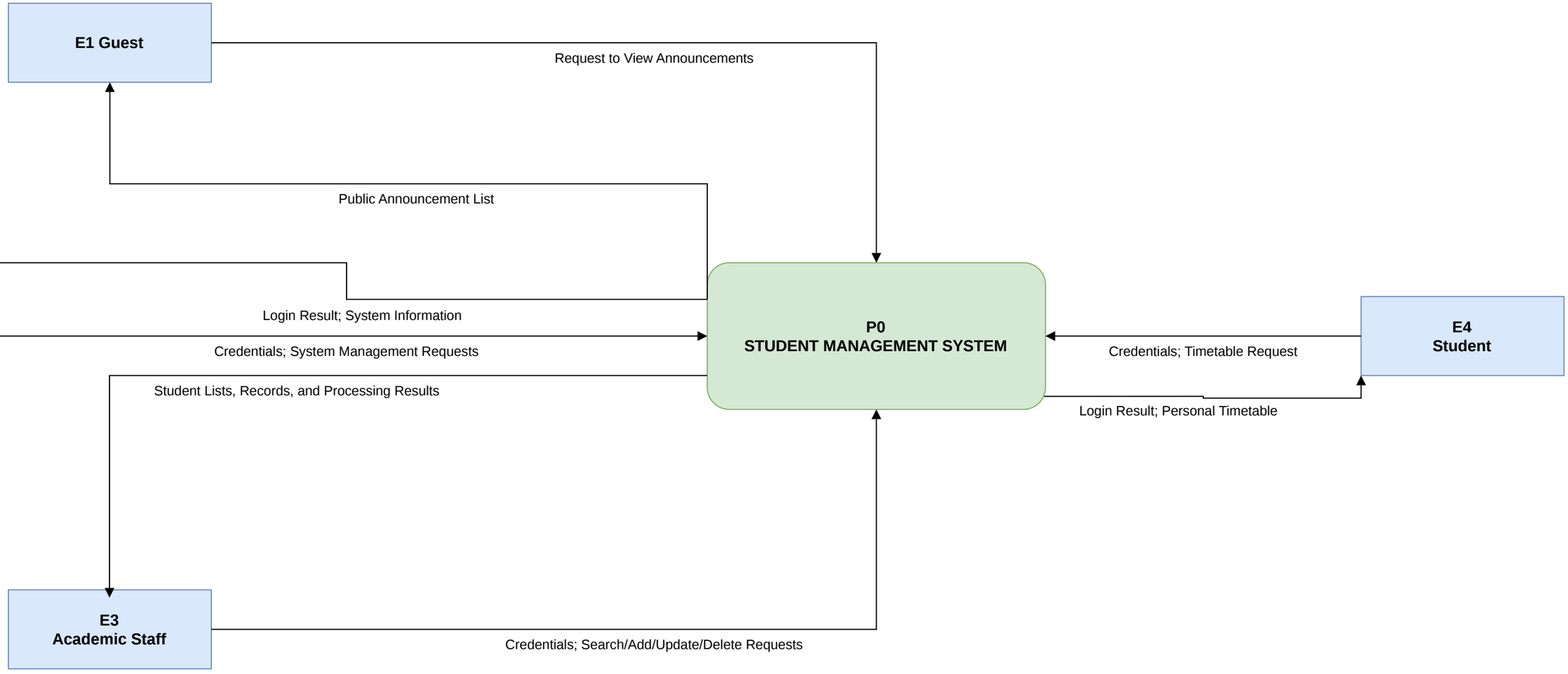
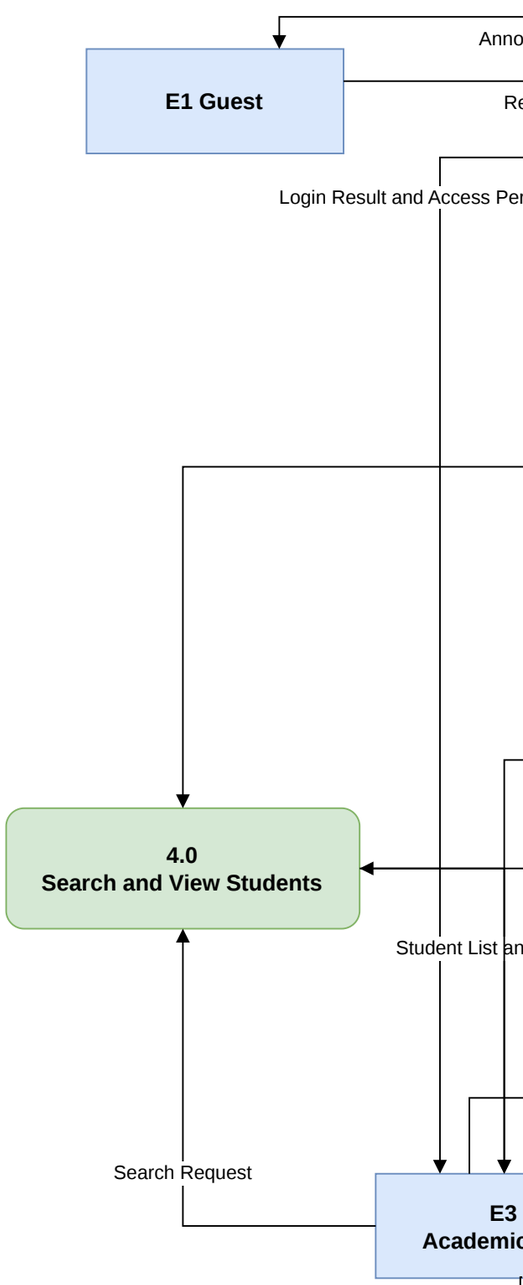

E2
Administrator


E1 Guest
Request to View Announcements
Public Announcement List
Login Result; System Information
Credentials; System Management Requests
Student Lists, Records, and Processing Results
P0
STUDENT MANAGEMENT SYSTEM
Credentials; Timetable Request
Login Result; Personal Timetable
E4
Student
E3
Academic Staff
Credentials; Search/Add/Update/Delete Requests

3.0
Manage Timetable


E1 Guest
Anno
Re
Login Result and Access Per
4.0
Search and View Students
Student List an
Search Request
E3
Academic

ouncements public

equest to View Announcements

**1.0**
**View Public Announcements**

**Manage Timetable**

Personal Timetable

**2.0**
**Login and Authorization**

Login Result and Access Permissions

Timetable Data

rmissions

Login Result and Access Permissions

Announcement List

Credentials and User Role

D2
Announcements

D1
User Accounts

D5
Timetable

D4
Classes and Majors

Class/Major Information

Timetable Request

**E2**
**Administrator**

Login Credentials

Class Information

**E4**
**Student**

Login Credentials

Updated Student Record

**5.0**
**Add and Update Students**

Add/Update Result

Existing Student Record

Student Information

D3
Student Records

Student Records

**6.0**
**Delete Student**

Student Record to Delete

nd Records

Valid Student Record Deletion

Result/Warning

Operation History

New/Updated Student Information

Related Records

**c Staff**

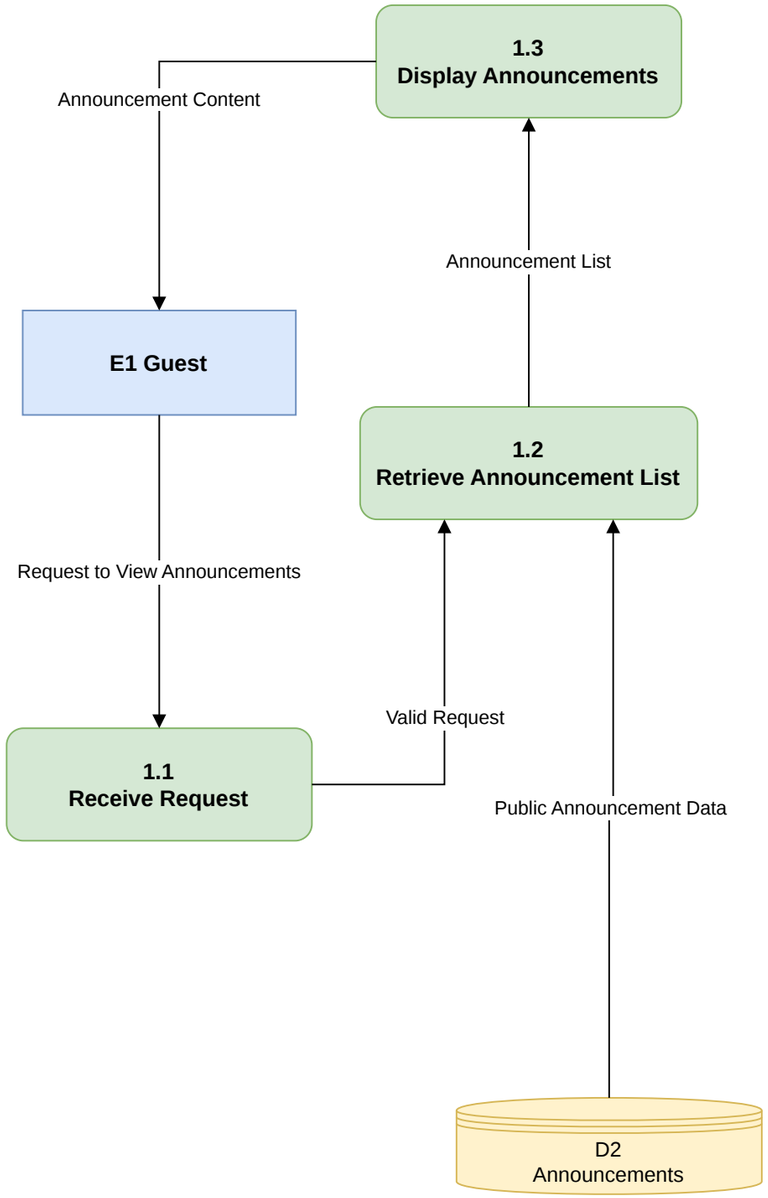
Login Credentials

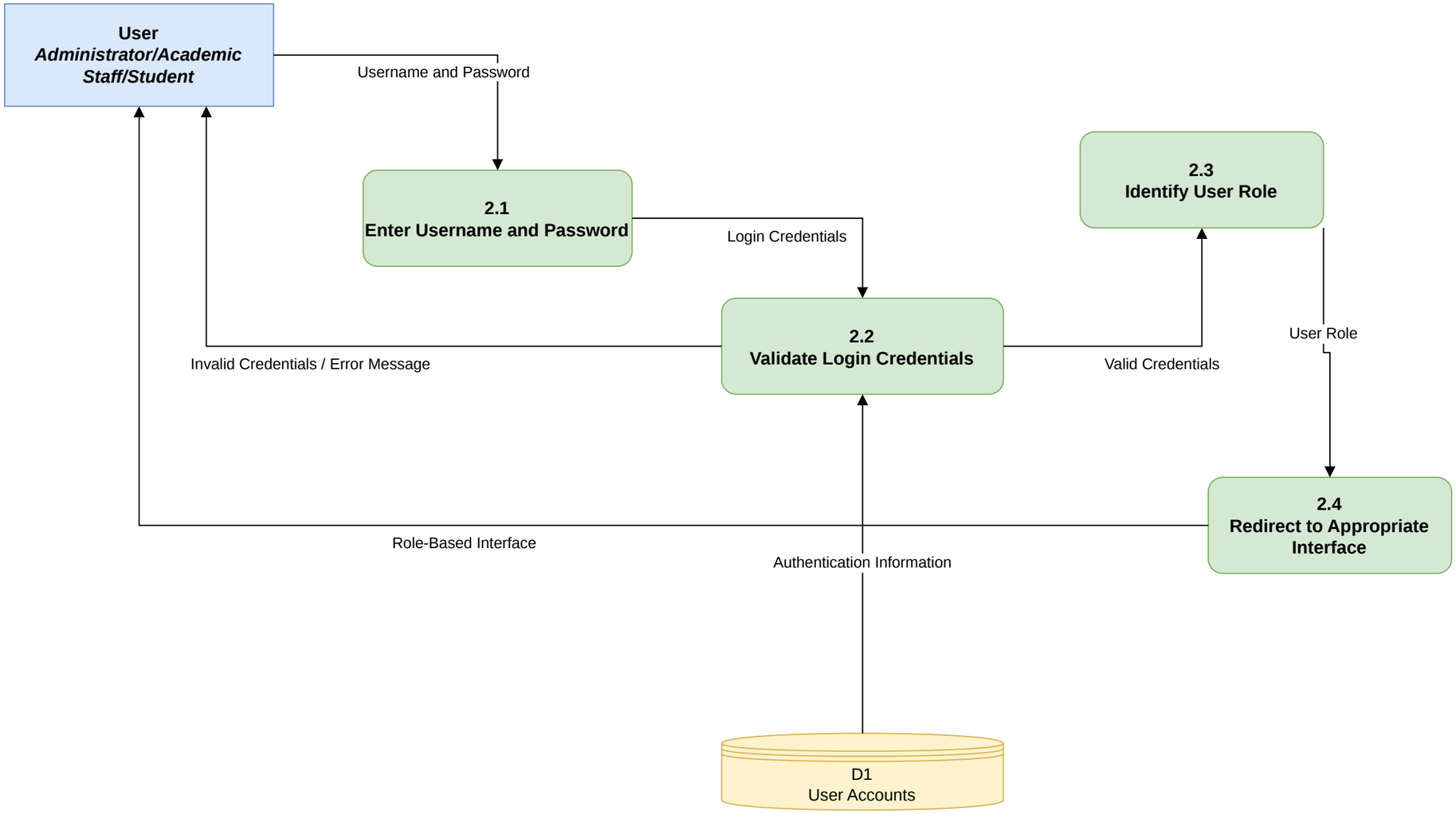
Deletion History

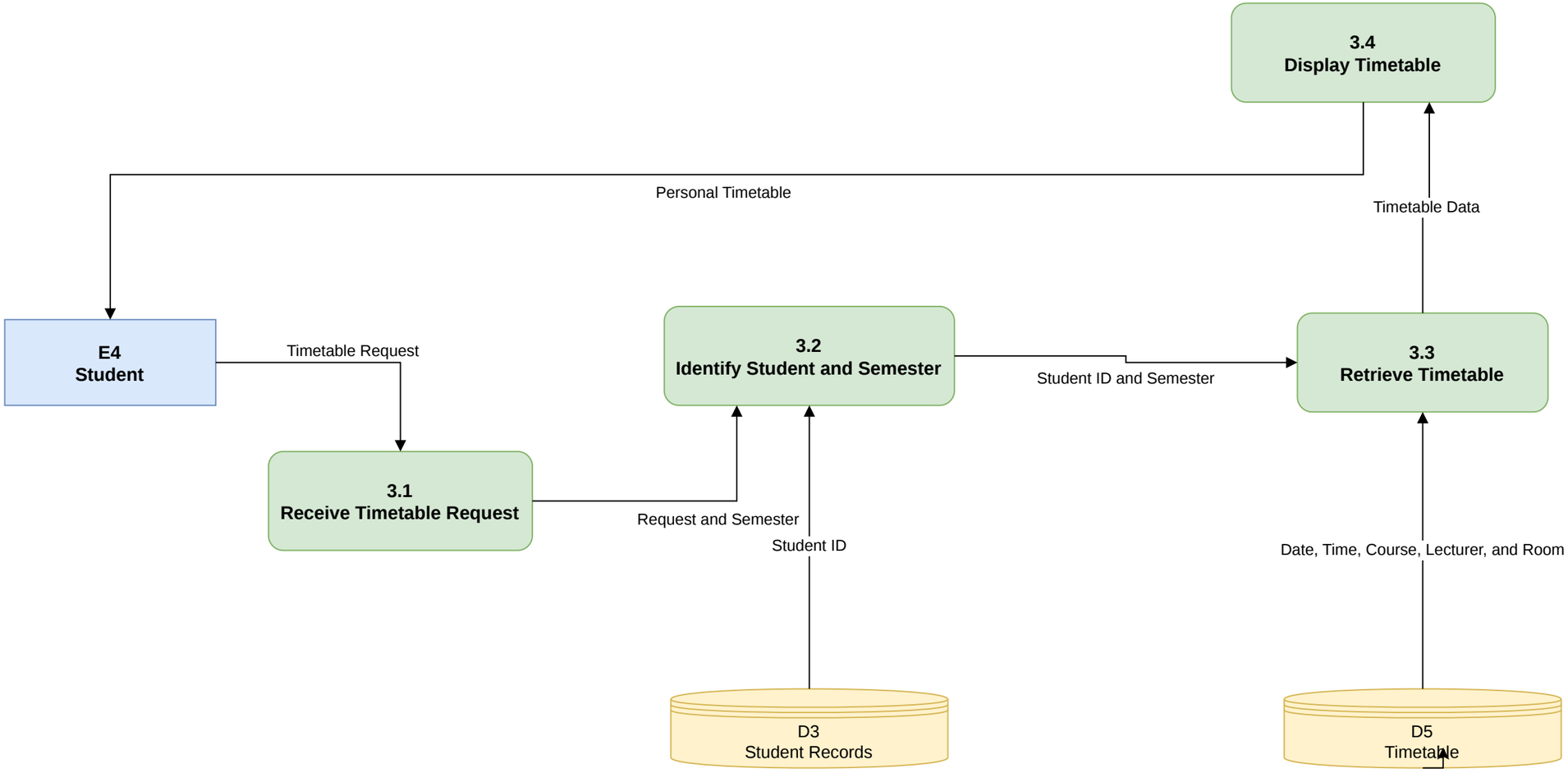
D6
Related Academic Records

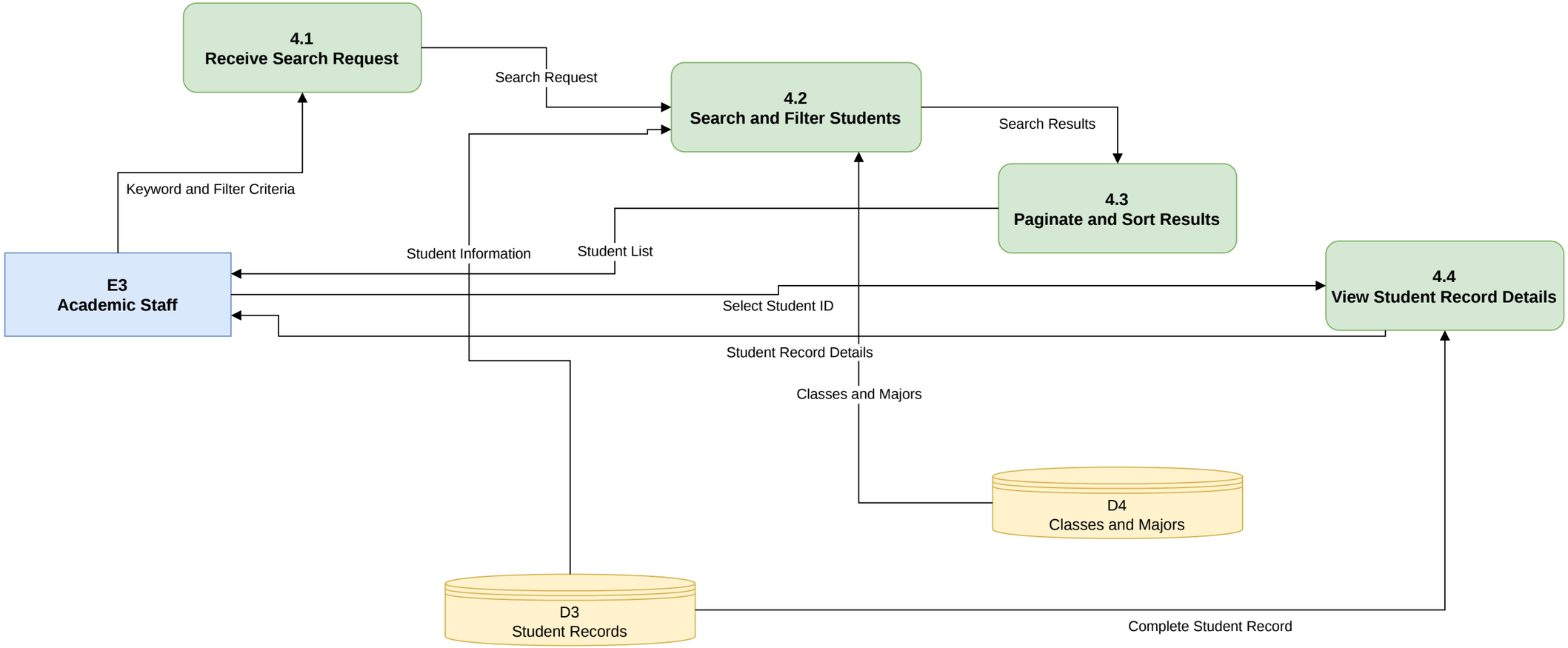
D7
System Audit Log

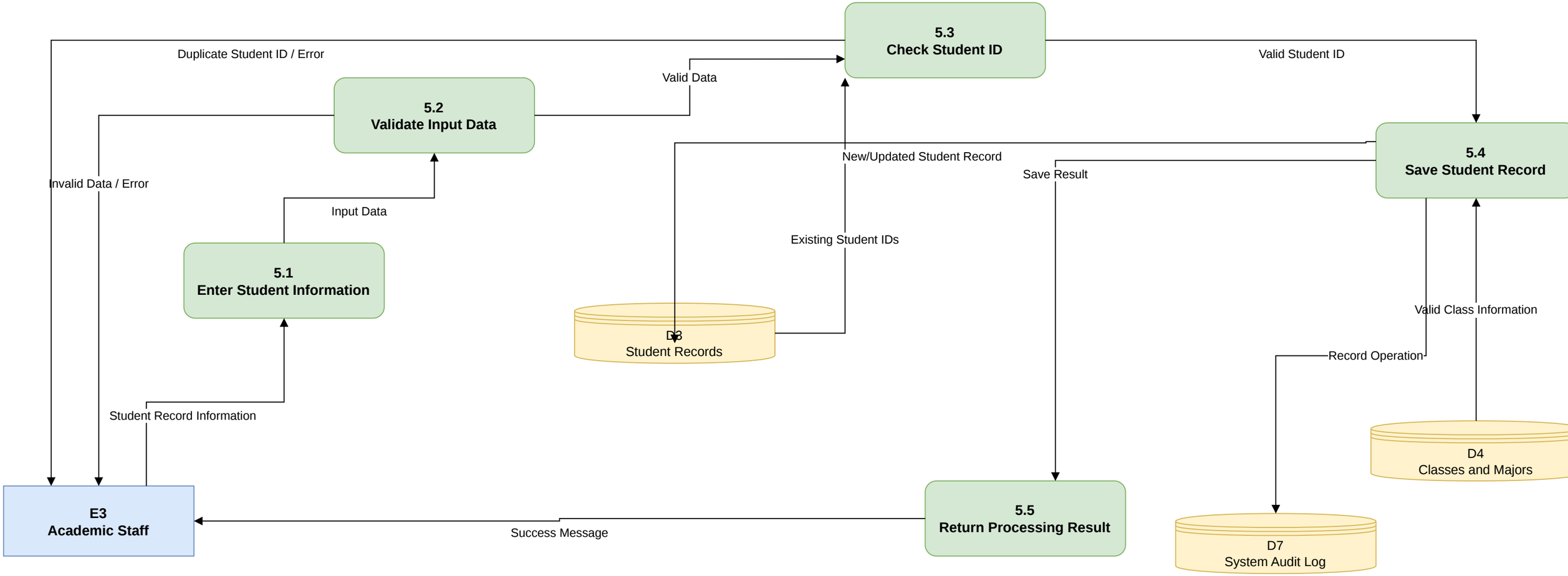
Delete Request and Confirmation


1.3
Display Announcements
Announcement Content
Announcement List
E1 Guest
1.2
Retrieve Announcement List
Request to View Announcements
Valid Request
1.1
Receive Request
Public Announcement Data
D2
Announcements

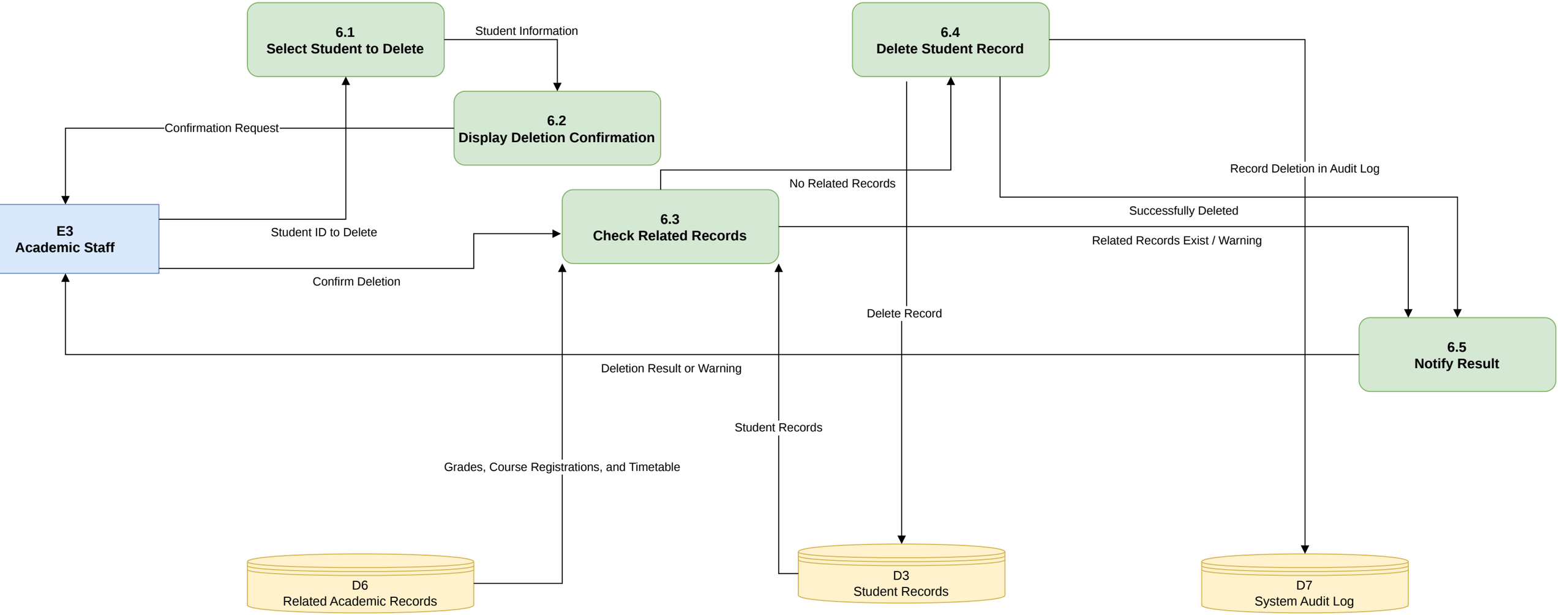

User
Administrator/Academic Staff/Student
Username and Password
2.1
Enter Username and Password
Login Credentials
2.2
Validate Login Credentials
Invalid Credentials / Error Message
Valid Credentials
2.3
Identify User Role
User Role
2.4
Redirect to Appropriate Interface
Role-Based Interface
Authentication Information
D1
User Accounts

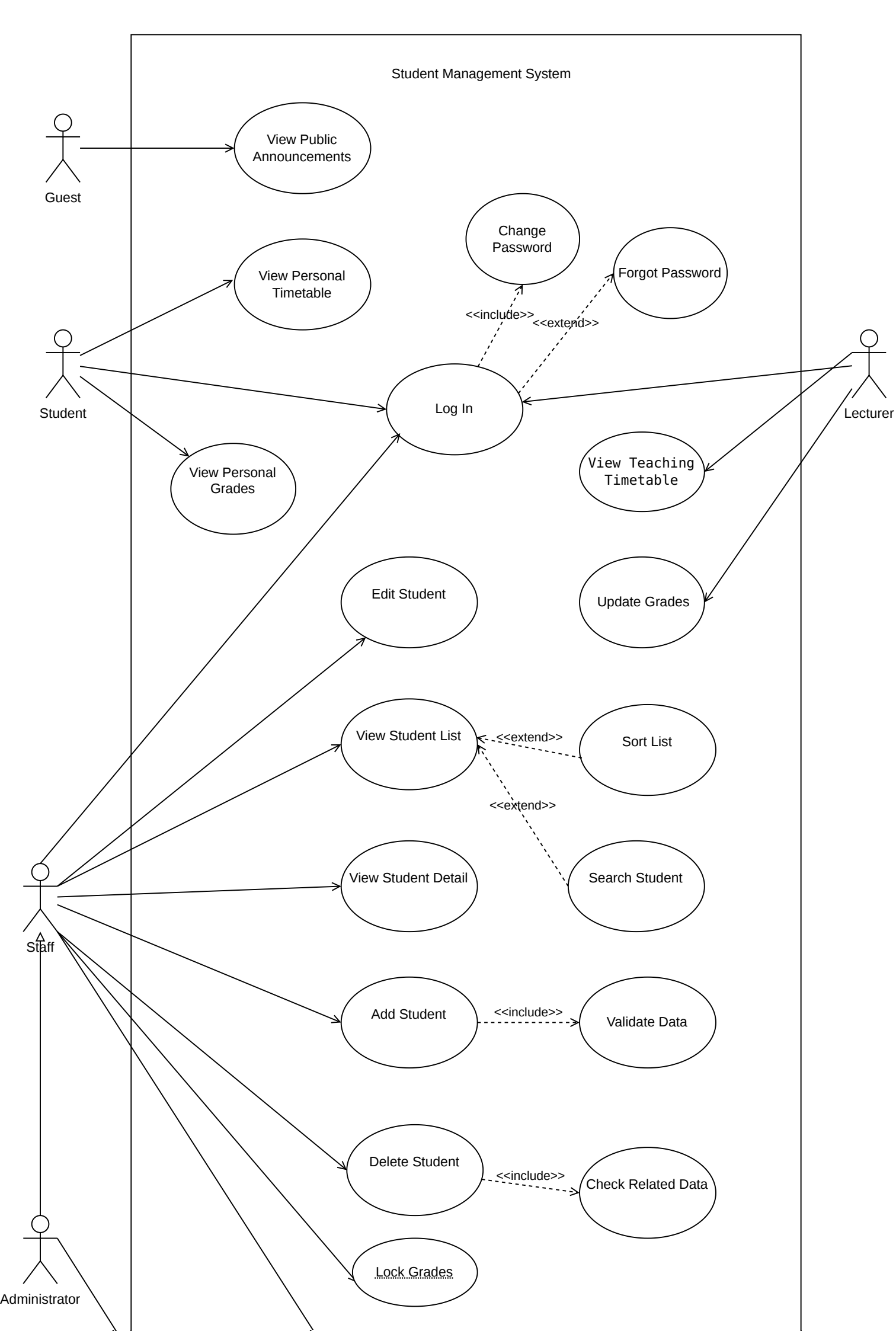

3.4
Display Timetable
Personal Timetable
Timetable Data
E4
Student
Timetable Request
3.2
Identify Student and Semester
Student ID and Semester
3.3
Retrieve Timetable
3.1
Receive Timetable Request
Request and Semester
Student ID
Date, Time, Course, Lecturer, and Room
D3
Student Records
D5
Timetable

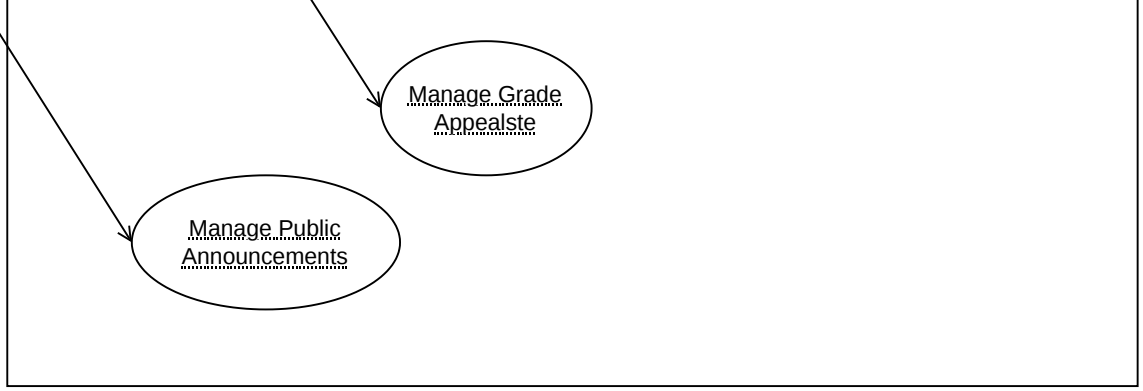

4.1
Receive Search Request
Search Request
4.2
Search and Filter Students
Search Results
4.3
Paginate and Sort Results
Keyword and Filter Criteria
Student Information
Student List
E3
Academic Staff
4.4
View Student Record Details
Select Student ID
Student Record Details
Classes and Majors
D4
Classes and Majors
D3
Student Records
Complete Student Record

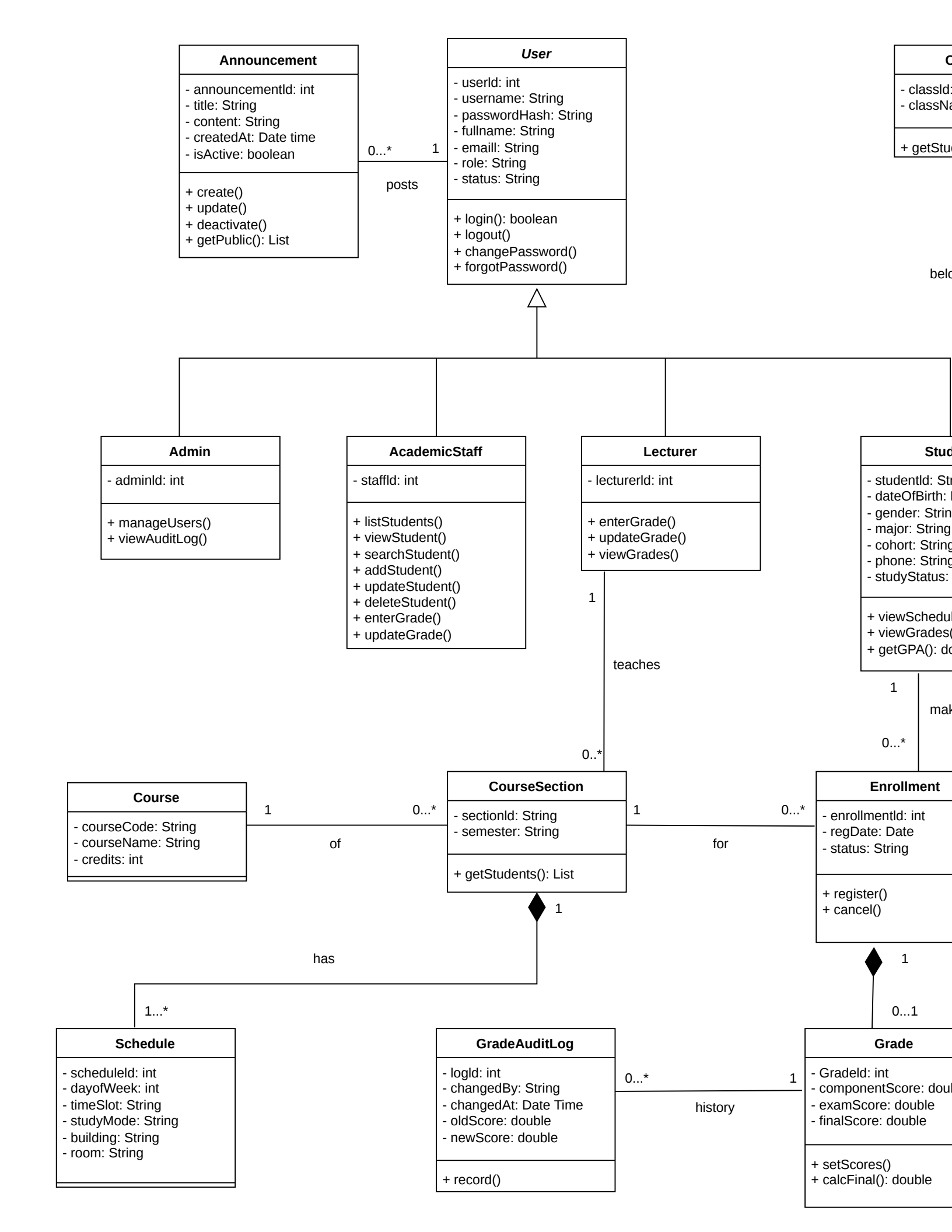

5.3
Check Student ID
Duplicate Student ID / Error
Valid Data
Valid Student ID
5.2
Validate Input Data
5.4
Save Student Record
New/Updated Student Record
Save Result
Invalid Data / Error
Input Data
5.1
Enter Student Information
Existing Student IDs
D3
Student Records
Valid Class Information
Record Operation
D4
Classes and Majors
Student Record Information
E3
Academic Staff
5.5
Return Processing Result
D7
System Audit Log
Success Message

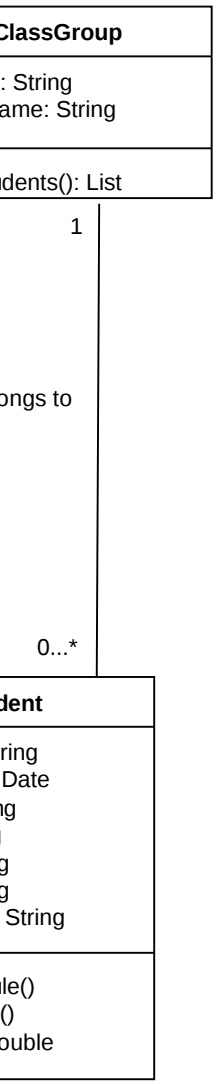



6.1
Select Student to Delete
Student Information
6.2
Display Deletion Confirmation
Confirmation Request
E3
Academic Staff
Student ID to Delete
Confirm Deletion
6.3
Check Related Records
No Related Records
6.4
Delete Student Record
Record Deletion in Audit Log
Successfully Deleted
Related Records Exist / Warning
Delete Record
6.5
Notify Result
Deletion Result or Warning
Student Records
Grades, Course Registrations, and Timetable
D6
Related Academic Records
Student Records
D3
Student Records
D7
System Audit Log

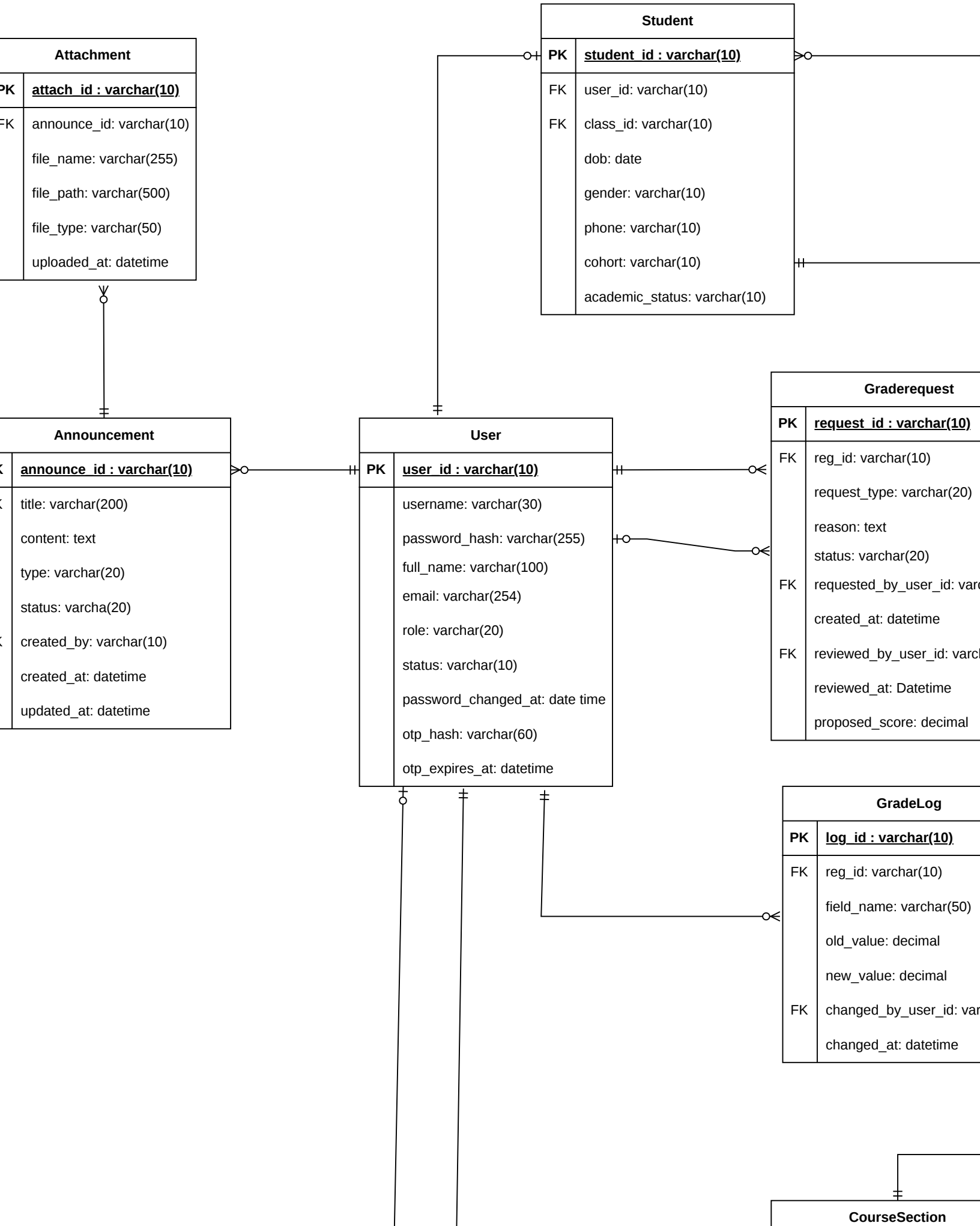

Student Management System
Guest
View Public Announcements
Change Password
Forgot Password
View Personal Timetable
<<include>>
<<extend>>
Student
Log In
Lecturer
View Personal Grades
View Teaching Timetable
Edit Student
Update Grades
View Student List
<<extend>>
Sort List
<<extend>>
View Student Detail
Search Student
Staff
Add Student
<<include>>
Validate Data
Delete Student
<<include>>
Check Related Data
Lock Grades
Administrator

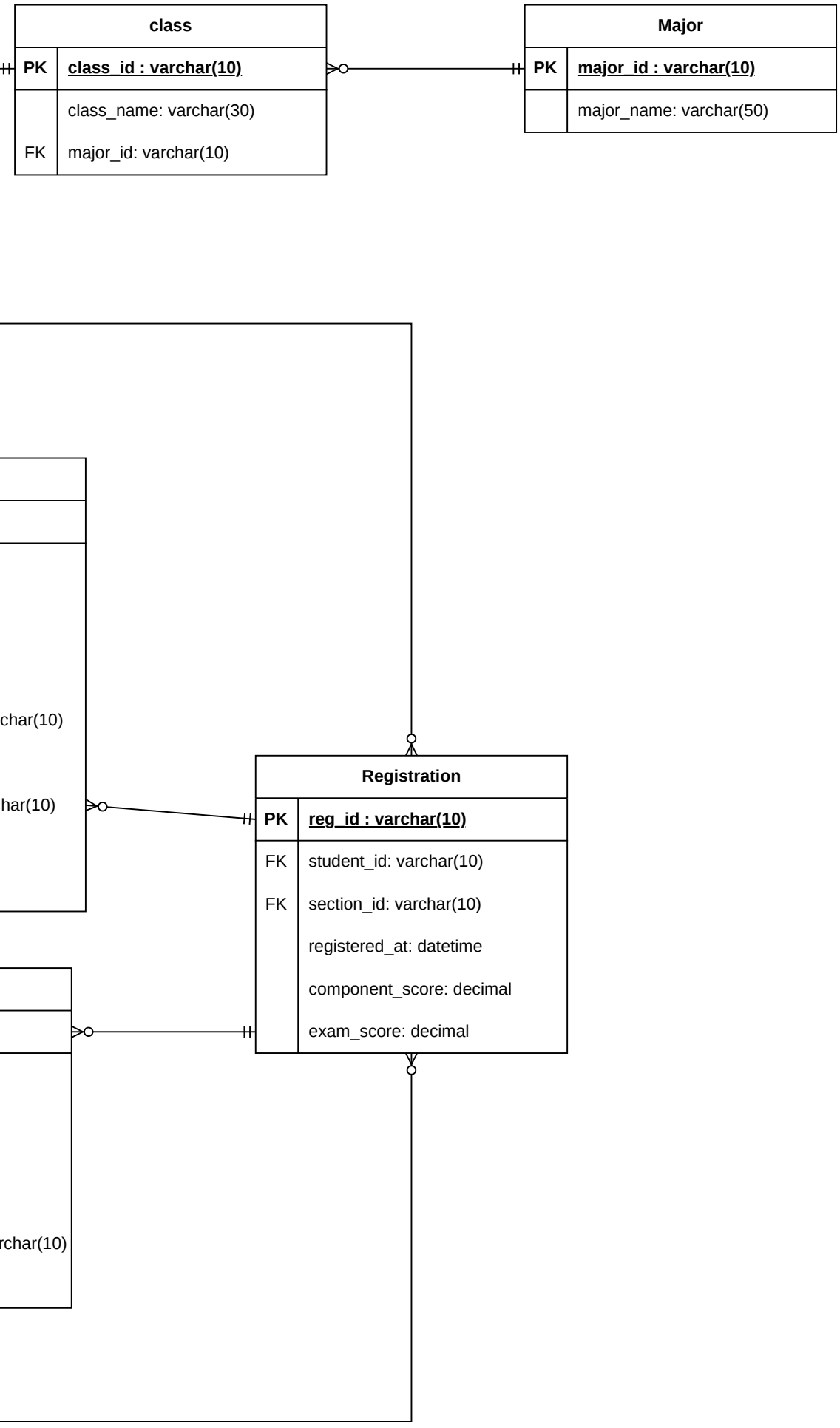

Manage Grade Appealste
Manage Public Announcements

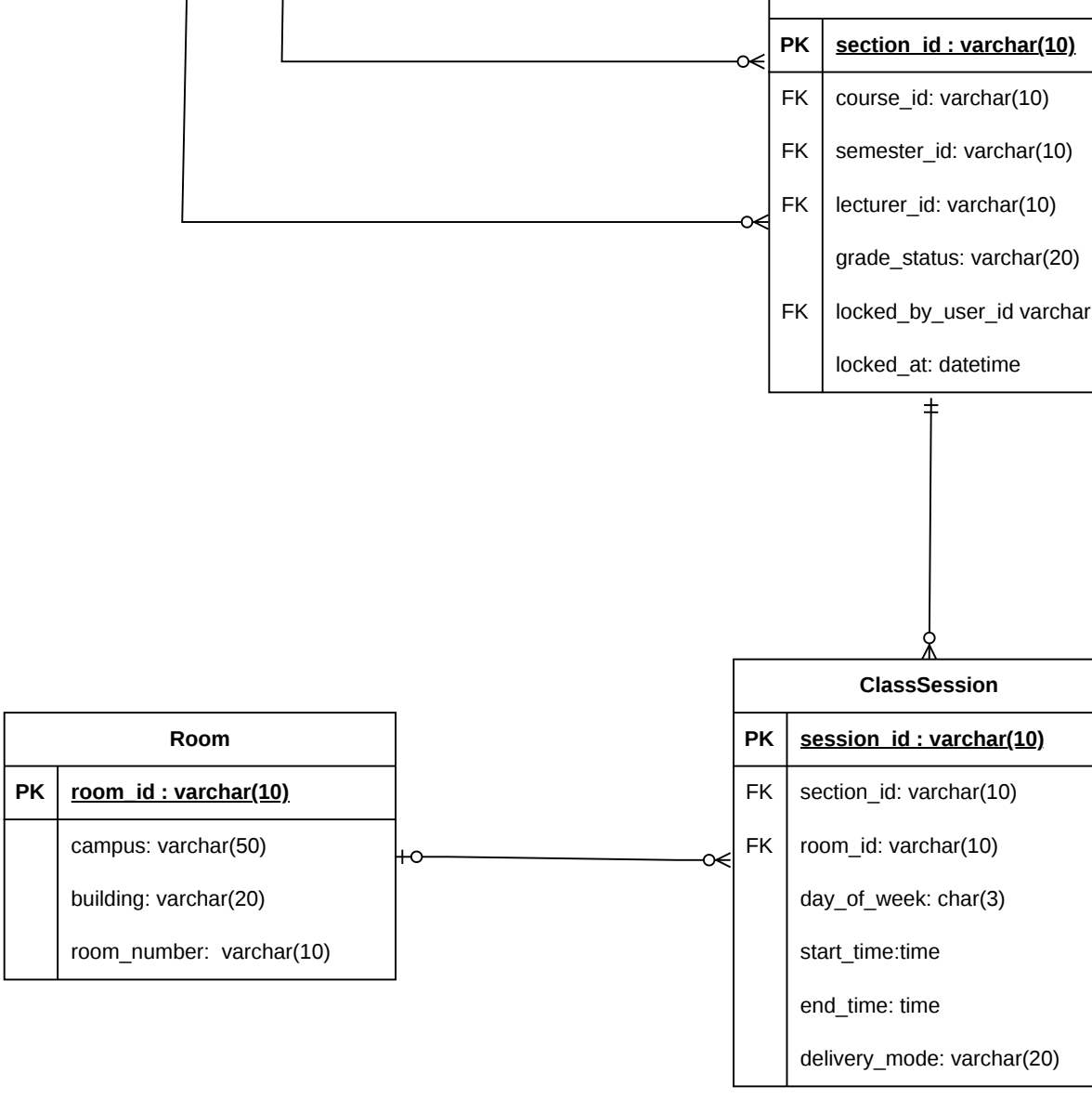

Announcement
- announcementId: int
- title: String
- content: String
- createdAt: Date time
- isActive: boolean
+ create()
+ update()
+ deactivate()
+ getPublic(): List
0...*
1
posts
User
- userId: int
- username: String
- passwordHash: String
- fullname: String
- emaill: String
- role: String
- status: String
+ login(): boolean
+ logout()
+ changePassword()
+ forgotPassword()
Admin
- adminId: int
+ manageUsers()
+ viewAuditLog()
AcademicStaff
- staffId: int
+ listStudents()
+ viewStudent()
+ searchStudent()
+ addStudent()
+ updateStudent()
+ deleteStudent()
+ enterGrade()
+ updateGrade()
Lecturer
- lecturerId: int
+ enterGrade()
+ updateGrade()
+ viewGrades()
1
teaches
0..*
Course
- courseCode: String
- courseName: String
- credits: int
1
0...*
of
CourseSection
- sectionId: String
- semester: String
+ getStudents(): List
1
0...*
for
Enrollment
- enrollmentId: int
- regDate: Date
- status: String
+ register()
+ cancel()
1
has
1...*
1
0...1
Schedule
- scheduleId: int
- dayofWeek: int
- timeSlot: String
- studyMode: String
- building: String
- room: String
GradeAuditLog
- logId: int
- changedBy: String
- changedAt: Date Time
- oldScore: double
- newScore: double
+ record()
0...*
1
history
Grade
- GradeId: int
- componentScore: doub
- examScore: double
- finalScore: double
+ setScores()
+ calcFinal(): double

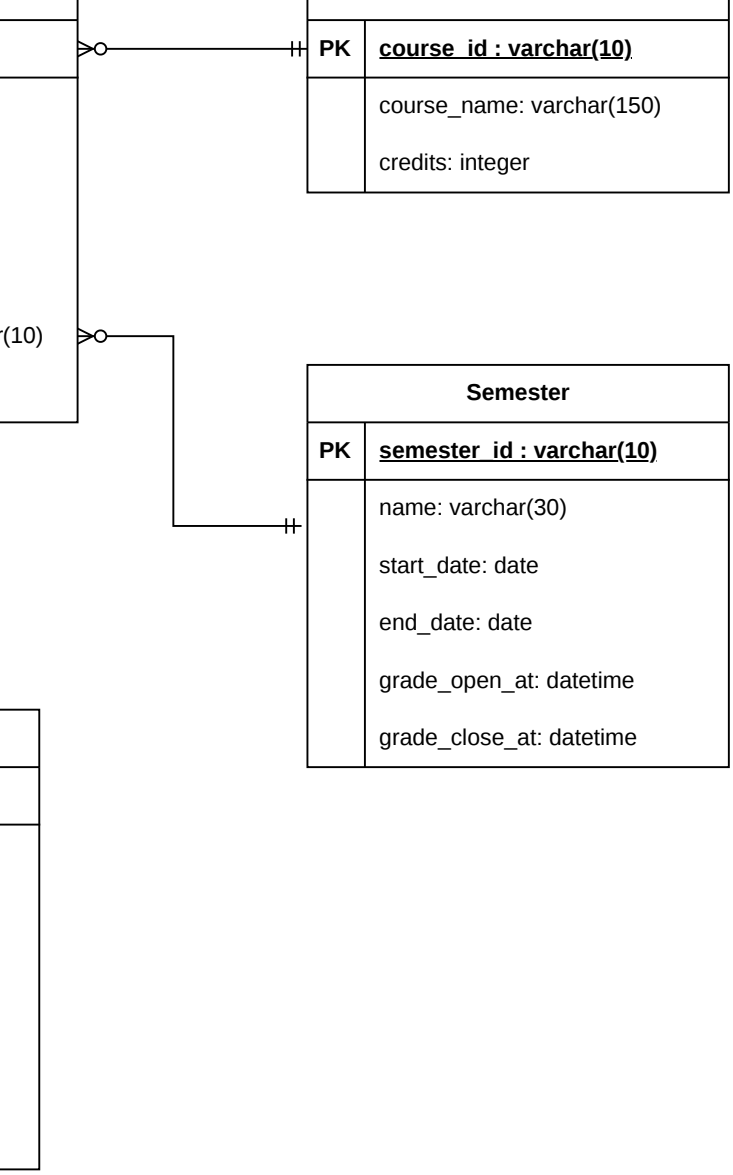

ClassGroup
: String
ame: String
dents(): List
1
ongs to
0...*
dent
ring
Date
g
g
g
String
le()
()
ouble
kes
ble




Attachment
PK
attach_id : varchar(10)
FK
announce_id: varchar(10)
file_name: varchar(255)
file_path: varchar(500)
file_type: varchar(50)
uploaded_at: datetime
Student
PK
student_id : varchar(10)
FK
user_id: varchar(10)
FK
class_id: varchar(10)
dob: date
gender: varchar(10)
phone: varchar(10)
cohort: varchar(10)
academic_status: varchar(10)
Announcement
announce_id : varchar(10)
title: varchar(200)
content: text
type: varchar(20)
status: varcha(20)
created_by: varchar(10)
created_at: datetime
updated_at: datetime
User
PK
user_id : varchar(10)
username: varchar(30)
password_hash: varchar(255)
full_name: varchar(100)
email: varchar(254)
role: varchar(20)
status: varchar(10)
password_changed_at: date time
otp_hash: varchar(60)
otp_expires_at: datetime
Graderequest
PK
request_id : varchar(10)
FK
reg_id: varchar(10)
request_type: varchar(20)
reason: text
status: varchar(20)
FK
requested_by_user_id: varc
created_at: datetime
FK
reviewed_by_user_id: varch
reviewed_at: Datetime
proposed_score: decimal
GradeLog
PK
log_id : varchar(10)
FK
reg_id: varchar(10)
field_name: varchar(50)
old_value: decimal
new_value: decimal
FK
changed_by_user_id: var
changed_at: datetime
CourseSection


class
PK
class_id : varchar(10)
class_name: varchar(30)
FK
major_id: varchar(10)
Major
PK
major_id : varchar(10)
major_name: varchar(50)
char(10)
har(10)
Registration
PK
reg_id : varchar(10)
FK
student_id: varchar(10)
FK
section_id: varchar(10)
registered_at: datetime
component_score: decimal
exam_score: decimal
rchar(10)
Course


PK
section_id : varchar(10)
FK
course_id: varchar(10)
FK
semester_id: varchar(10)
FK
lecturer_id: varchar(10)
grade_status: varchar(20)
FK
locked_by_user_id varchar
locked_at: datetime
ClassSession
PK
session_id : varchar(10)
FK
section_id: varchar(10)
FK
room_id: varchar(10)
day_of_week: char(3)
start_time:time
end_time: time
delivery_mode: varchar(20)
Room
PK
room_id : varchar(10)
campus: varchar(50)
building: varchar(20)
room_number: varchar(10)


PK
course_id : varchar(10)
course_name: varchar(150)
credits: integer
(10)
Semester
PK
semester_id : varchar(10)
name: varchar(30)
start_date: date
end_date: date
grade_open_at: datetime
grade_close_at: datetime